Anh Dao

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EXPERIENCE

Smartly – ***Software Engineer*** ***Feb 2026 – Present***

- Working on the Agentic Orchestration team for Smartly's core initiative Synapse
- Building Smartly's **Pydantic AI**-based agentic orchestration harness and Smartly's external MCP server
- Maintaining **PostgreSQL** storage and **Redis** caching for agents' metadata, message history and draft outputs, plus PoCs for a persisted artifacts log
- Prototyped the company's 1st external MCP gateway and **TypeScript**-based MCP servers, creating the company guidelines for good tool design and MCP implementation
- Assisted in the 2-person migration of the company's Frontend monorepo of 4 million lines of TypeScript from **Jest** to **Vitest** and 4 more repos from Jest/ESLint/Prettier to Vitest/Oxlint/Oxfmt solo
- Optimized 2 existing LLM-based services through **LangFuse** trace monitoring and evals-based prompt optimization, reducing the cost of inference by 2.5x without loss of performance
- Part of the company's AI Productivity Frontier group, delivering Knowledge Sharing sessions to 200+ R&D employees

Musixmatch – ***Research Assistant*** ***Feb 2026 – Aug 2026***

- Collaborated with Professor Eric Malmi of Aalto University to build an ETL data pipeline in Python to exhaustively collect, clean, and deduplicate Musixmatch's lyrics dataset through their provisioned API, up to 10M+ tracks
- Ran batch LLM annotation on 15,000+ Finnish tracks, cleaning up malformed LLM outputs into structured data
- Replicated prior research on Finnish Rhyme analysis, extending it to other genres and languages. Link

Nokia Solutions & Networks – ***R&D Trainee*** ***Aug 2025 – Jan 2026***

- Built a pipeline indexing 100,000+ documents into **Vertex AI Search** on **GCP**, unifying product search for RAG
- Prototyped a multi-agent harness with Gemini Agent Platform combining fanout research, network logs triaging, and RAG-based QnA, reducing the incident triage time by ∼40%

Aalto University – ***Teaching Assistant & Master's Thesis Worker*** ***Aug 2024 – Jul 2026***

- Assisted in core Master's-level Security courses, totalling over 500 students, by holding office hours, designing network security assignments, and creating tools for fast grading
- Thesis: gradient attacks and a confidence-triggered fallback defense for VR cybersickness LSTM classifiers

PROJECTS

Assorted Agentic Systems Projects ***2026 – Present***

- Participating in Teleport Coding Challenge, porting Nethack 5.0, a complex roguelike game with 500,000 lines of C and Lua code to JavaScript with byte-exact behavior
- Designing guardrails for multi-agent orchestration, multi-turn evals framework to prevent common pitfalls with independent long-horizon agents like reward hacking, challenge bypass and goal misalignment
- Building a personal assistant on Cloudflare Durable Objects using the Pi agent harness, extended with custom OptChat-based context management

Scalable Agentic Systems Evaluation ***Jan – Apr 2026***

- Benchmarked 300 software engineering tasks (SWE-Bench Lite) across 3 different harnesses and 6 open weight models, deployed on CSC's LUMI supercomputers with **SLURM** and **ROCm**
- Profiled GPU utilization and evals results, fixing incompatibilities with LUMI's **Singularity**-based Containers, models' malformed reasoning, and erroneous tool usage

EDUCATION

M.Sc. (Tech.) in Computer Science, Aalto University (GPA 4.09/5.00, ML minor) ***Aug 2024 – Jul 2026***

B.A. in Computer Science, Grinnell College (GPA 3.76/4.00) ***Aug 2019 – May 2023***

SKILLS

Languages: TypeScript, Python, Go, Java, C/C++, SQL, R
Frameworks & Libraries: Node.js, React, Jotai, Pydantic AI, MCP, PyTorch, vLLM, MLX
Tools & Platforms: PostgreSQL, Redis, Kafka, BullMQ, Kubernetes, AWS/GCP, LangFuse, Grafana, Playwright